# Big Data Analytics & Predictive Intelligence

***Customer Transactions — E-commerce Domain***

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## 1. Abstract

This project implements a complete big-data analytics pipeline for an e-commerce customer transactions dataset, covering data preprocessing, SQL-based business intelligence, customer segmentation via RFM analysis, a predictive churn model, and an executive dashboard. The pipeline processes 4,000 transactions across 1,148 customers, identifying key revenue drivers and a churn rate of 29.5%, with a Random Forest classifier achieving an ROC-AUC of 0.708 in predicting customer churn.

## 2. Objectives

• Collect and process a large-scale e-commerce transactions dataset

• Perform advanced feature engineering using RFM (Recency, Frequency, Monetary) analysis

• Use SQL to answer key business intelligence questions

• Implement a machine learning model for churn prediction

• Build an executive dashboard summarizing KPIs and model results

• Generate business recommendations from the analysis

## 3. Dataset Description

A synthetic but realistic e-commerce transactions dataset was generated, simulating 18 months of activity (Jan 2025 – Jul 2026) across 8 major Indian cities and 7 product categories. The dataset includes 4,000 transactions from 1,148 unique customers, with fields for transaction amount, category, payment method, customer demographics, and returning-customer status.

| Property | Value |
|---|---|
| Total Transactions | 4,000 |
| Unique Customers | 1,148 |
| Date Range | Jan 2025 – Jul 2026 |
| Cities Covered | 8 |
| Product Categories | 7 |

| | |
|---|---|
| Total Revenue | ~Rs 16.3M |

## 4. Methodology

**Feature Engineering:** For each customer, Recency (days since last purchase), Frequency (number of transactions), and Monetary (total spend) values were computed, along with average order value, category diversity, and demographic attributes.

**Churn Definition:** A customer was labeled as churned if their most recent transaction occurred more than 180 days before the dataset's snapshot date.

**Model:** A Random Forest Classifier (200 trees, max depth 8, balanced class weights) was trained on an 80/20 train-test split using frequency, monetary, average order value, unique categories, age, city, and gender as features.

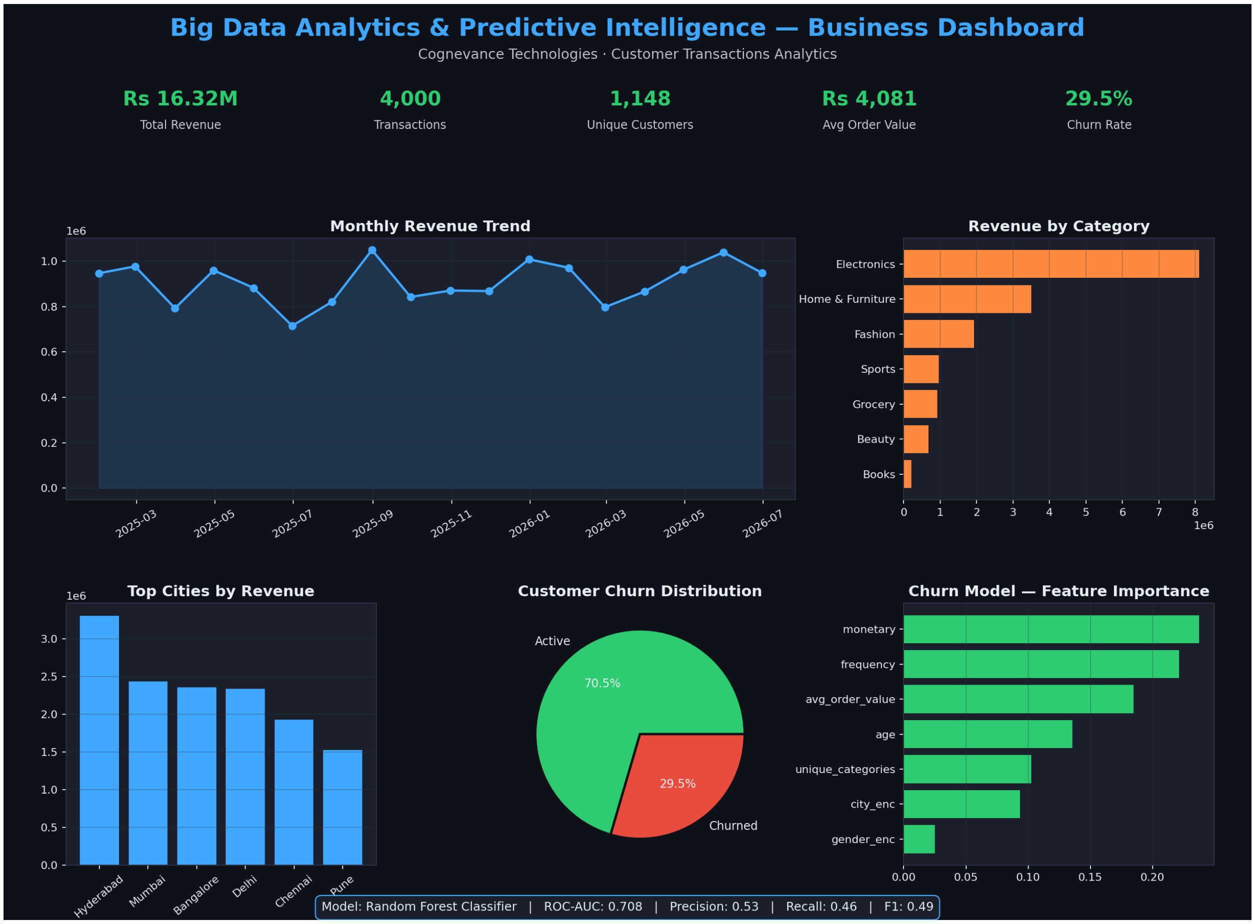
## 5. Results

| Metric | Value |
|---|---|
| ROC-AUC Score | 0.708 |
| Precision (Churn class) | 0.53 |
| Recall (Churn class) | 0.46 |
| F1-Score (Churn class) | 0.49 |
| Overall Accuracy | 0.72 |

The top predictors of churn were **total monetary spend** and **purchase frequency**, followed by average order value and customer age — indicating that low-engagement, low-spend customers are most at risk of churning.

## 6. Dashboard

An executive dashboard was built summarizing key KPIs (revenue, transactions, unique customers, average order value, churn rate), monthly revenue trends, category and city performance, churn distribution, and model feature importance.




## 7. Business Recommendations

• Launch targeted re-engagement campaigns for the 29.5% churn-risk segment, focusing on low-frequency, low-monetary customers.

• Prioritize inventory and marketing investment in Electronics and Home & Furniture, the top revenue-driving categories.

• Replicate Hyderabad and Mumbai's performance in lower-revenue cities through localized promotions.

• Introduce a loyalty/rewards program targeting mid-frequency customers to increase retention before churn occurs.

## 8. Conclusion

This project demonstrates an end-to-end big data analytics workflow — from raw transaction data to a deployable churn prediction model and an executive-ready dashboard. The RFM-based feature engineering combined with a Random Forest classifier provides actionable insight into customer churn risk, directly supporting data-driven retention strategy.